005930.KS - Samsung Electronics

Full investment research | Model assumptions and source evidence

Framework 8.0 | Q3 preliminary earnings guidance incorporated October 11, 2026 (Toronto) | HBM operating path retained | Architecture evidence reviewed October 6

HELD. Analytical Base: 50/60 = 83/100. Portfolio Excel Base: 83/100.

Active Q4 2027 ceiling: KRW 641,449.06 per share. Q1 2028: KRW 643,911.55. Valuation remains 6x forward-four-quarter operating profit.

The model now separates HBM, conventional DRAM and NAND. One active case includes a **50% probability-weighted increment** for HBM pricing/mix, together with its production, packaging, yield and operating-cost burden. No second probability haircut is applied to the ceiling. The existing NAND +23%/+20%/+15% calendar path, conventional DRAM residual without further repricing, non-memory businesses, 6.648649811B economic-share denominator and scoring are retained. Currency is KRW trillions unless stated. Q3 2026 consolidated revenue and OP now use issuer preliminary guidance; its segment splits remain model estimates. Q4 2026 onward remains modeled, not reported results.

The HBM source note supplied by the owner was dated October 1 in Korea and incorporated September 30 in Toronto. That historical HBM update is distinct from this October 11 incorporation of the October 8 Q3 preliminary guidance. No future-period forecast is presented as an observed result.

October 11 earnings update: Samsung's October 8 preliminary Q3 2026 guidance is **KRW 195T revenue and 107.4T operating profit**, implying a **55.08%** consolidated margin. These are management estimates ahead of final results, not audited actuals. The prior model was 187.586027357573T revenue and 103.138027357573T OP. The difference remains an **unallocated consolidated reconciliation**, not an imputed increase to Memory. The active **Q4 2027 ceiling stays KRW 641,449.06**; only the already-past Q3 2026 valuation window changes to **KRW 477,872.43**. [SAMS-Q3G]

1. Thesis, products and competitive position

HELD. BOTTLENECK in memory supply, owned through a diversified electronics company. HBM and conventional server memory drive most modeled profit. Foundry, packaging and devices expand technology exposure but also create distinct execution and capital-allocation demands. The 6x company-specific operating-profit multiple remains sanctioned and unchanged.

The bottleneck and the whole company

Samsung combines a large memory manufacturing franchise with foundry/system semiconductor activities, mobile devices, displays and consumer electronics. HBM and advanced memory can possess scarce capacity and qualification advantages while the consolidated business still reflects unrelated product cycles, internal transfers and capital allocation. This is the framework's Rule 10 issue: the attractive niche must be translated into its actual contribution to the whole company.

The approved four-vector operating build separates DS Memory, DS LSI + Foundry, DX net-sales proxy and Other net-sales proxy. HBM, conventional DRAM and NAND are economically distinct within DS Memory, and now have separate modeled revenue, cost and operating-profit schedules. The split is a calibrated proxy; Samsung has not disclosed a complete product-level income statement. HBM requires qualified designs, advanced stacking and packaging yields as well as wafer supply. Conventional DRAM and NAND face different mix and supply cycles. A packaging investment is an enabling input, not an additional sale of a chip already

counted in memory revenue. Internal memory supplied to another Samsung business cannot be added again to consolidated revenue without the appropriate elimination.

Scale, competition and attribution

Manufacturing scale, process expertise, a broad customer base and financial resources support the user's mega-cap conviction. They do not remove the need to validate HBM qualifications, foundry utilization and the profitability of the non-memory businesses. Samsung, SK hynix and Micron compete on technology, yield, customer qualifications and allocation; a market-wide shortage does not establish a single supplier's product share.

Share-class attribution matters because preferred shareholders also participate in economic earnings. The approved bridge below uses common and preferred economic claims and the relevant treasury-stock treatment. It is not appropriate to allocate all consolidated operating profit solely to the listed common-share count. The analytical score remains visible as a recommendation; the portfolio Excel Base is 83 following the owner's October 7 removal of the old override. This edition corrects stale 99 text; it does not change today's Excel Base.

Research provenance: the product and competitive analysis in this section synthesizes the supplied original research. Dated market-share, process and channel claims retain their original evidence status; this integration is not a new independent verification of those claims. Current earnings and model assumptions are set out below.

Integrated co-design and long-run memory substitution

Samsung's combination of DRAM, logic process, interface engineering and packaging offers a plausible co-design advantage: the supplier can optimize both ends of the memory interface and understand package/process variation earlier. That is a mechanism to test through accepted output and production cost, not proof of a permanent lead over every competitor. SK hynix and Micron can also co-design with foundry and interface partners; ownership of every step is not the only route.

Samsung's February 12, 2026 commercial HBM4 announcement specifies 1c DRAM, a 4 nm logic base die and consistent 11.7 Gbps operation, with capability up to 13 Gbps. It predates the article's May 31 assertion that all vendors failed the 11 Gbps requirement. The announcement is company-reported evidence and does not supply a complete yield distribution. The source's SF4X-versus-TSMC and Samsung-versus-other-SiPh rankings lack matched product conditions and commercial-cost data, so they do not create an automatic market-share transfer.

The package measurements reproduced below show why channel characterization matters. They concern the path between an ASIC and HBM's base die; they do **not** measure the internal base-die-to-DRAM stack or disclose Samsung's hidden stack parasitics.

HBM Integration Challenges: Signal and Power Integrity

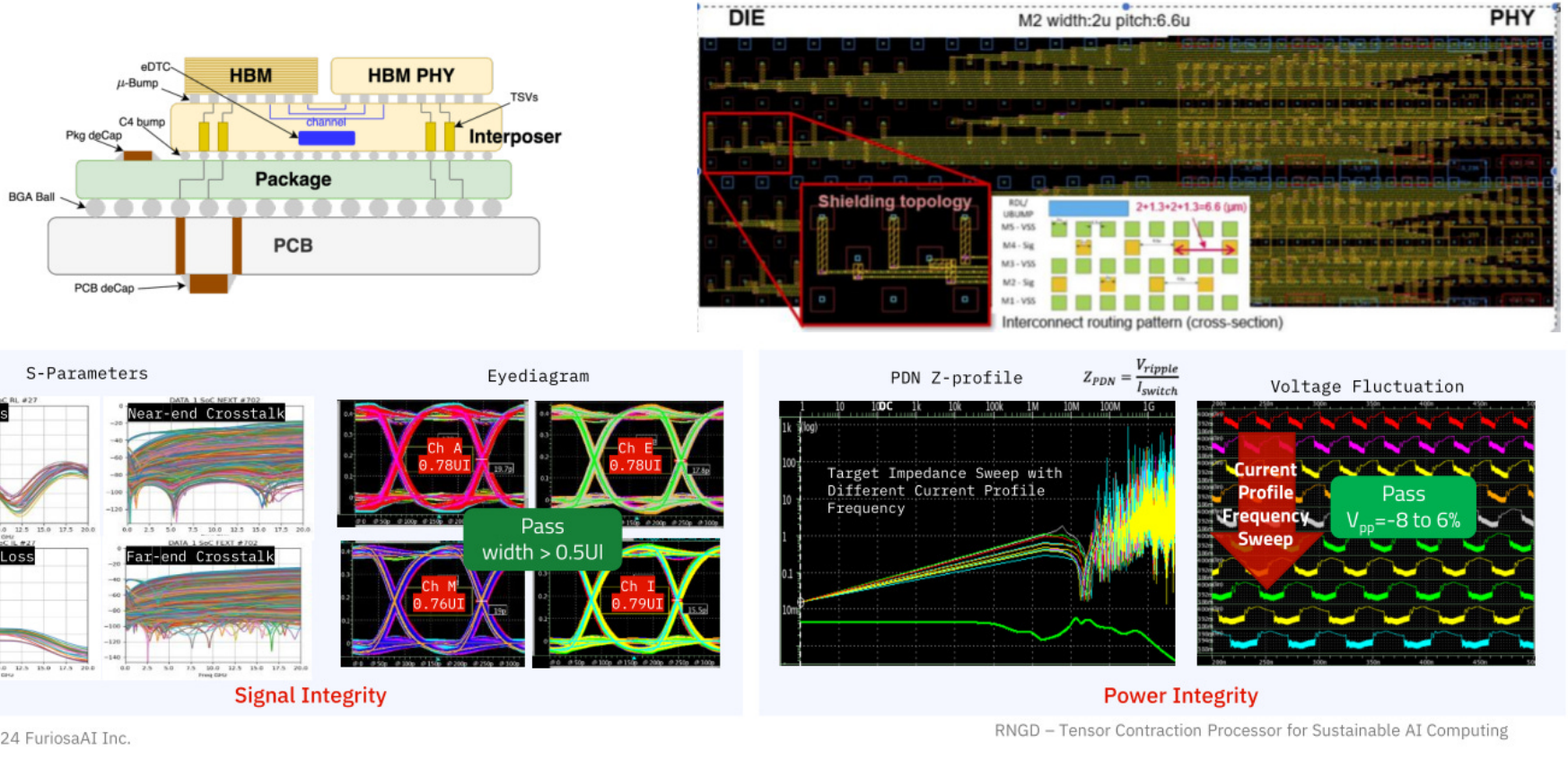


Figure IA24-S1. Furiosa AI Hot Chips 2024 package signal/power-integrity slide reproduced in IA24, body image 113. This is another company's package example, not Samsung qualification data or internal HBM-stack measurement.

The existing HBM-01 model already weights the pricing/mix outcome at 50% and includes front-end cost +10%, packaging cost +20%, a 95% to 90% yield stress proxy and additional R&D/SG&A equal to 5% of incremental sales. These are disclosed model assumptions, not measured Samsung yields. The older integration thesis supplies no additional qualified units or independently measurable cost advantage beyond that branch. Internal foundry transfers also do not create another consolidated profit pool. The HBM premium fades in 2029 and disappears in 2030 under the existing assumption. No extra margin premium, volume shock or Base increase is added.

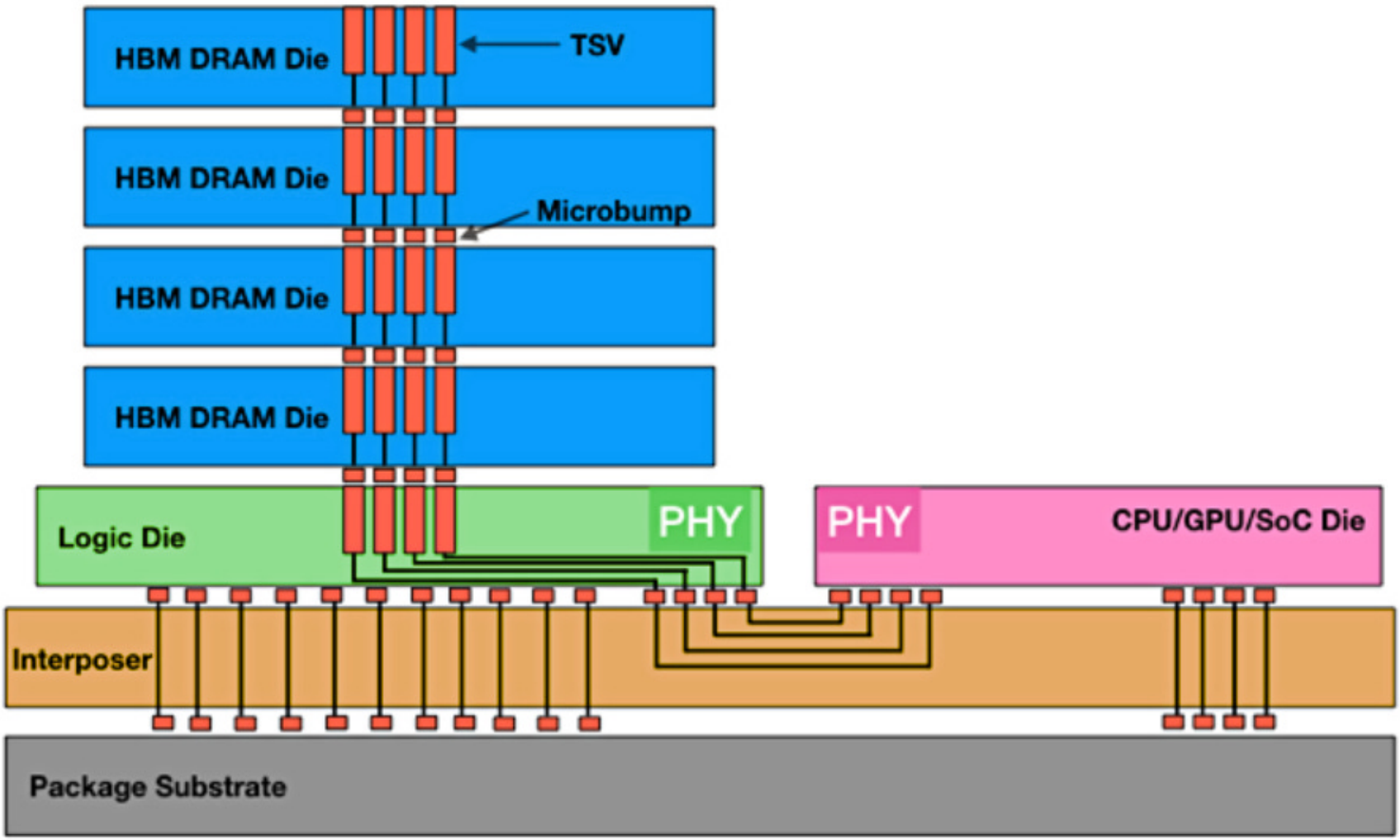


Figure IA24-S2. HBM/base-die/interposer structure reproduced in the May 31 article, body image 5. It maps which production steps a co-design hypothesis must improve; the illustration is not a cost forecast.

Optical memory: promising link physics, incomplete system economics

The May 31, 2026 Irrational Analysis article argues that clock-forwarded optics could eventually connect accelerators to reusable LPDDR pools. HBM is a DRAM product architecture; losing HBM attachment does not imply that its supplier loses every displaced memory bit. Conversely, selling more conventional bits does not guarantee the same revenue or margin. Custom HBM, local HBM caches, LPDDR/SOCAMM and pooled memory can coexist across workloads.

NVIDIA's ISSCC 2026 research supports the link mechanism, not a shipped disaggregated-memory platform. Its supplied comparison table retains thermal-tuning and laser-efficiency assumptions. Link energy and interface density do not measure complete memory-access latency, useful application bandwidth, controller cost or software utilization. The author's NVIDIA 2028-2029 and Broadcom/Marvell 2029-2031 timing is a forecast, not a product commitment. A clock-forwarded link removes some electronics; optical propagation, controller access and resource allocation still take time. CXL pooling is a separate protocol and controller architecture, not a synonym for an optical memory pool.

Performance Comparison

		This work	VLSI'22 [1]	VLSI'23 [9]	ISSCC'23 [3]
Process	EIC	7nm FinFET	45nm SOI CMOS	28nm CMOS	7nm CMOS
	PIC	65nm SOI SiPh		300mm SOI SiPh	45nm SOI SiPh
Integration		Hybrid	Monolithic	Hybrid	Hybrid
Architecture		FC	EC	EC	EC
Number of wavelengths		9	8	8	7
Per-λ Data rate <Gbps>		32	16	32	50
Per-Fiber Data rate <Gbps >		256	128	256	350
TX+RX Lane size <mm²>		0.013	-	~1	0.031 (RX only)
EIC shoreline density[a] <Tbps/mm>		**0.8**	~0.14	~0.064	-
EIC area density[b] <Tbps/mm² >		**1.33**	~0.064	~0.032	-
RX sensitivity OMA <dBm>		**<-16.0**[c]	-	-10	-10.1 to -11.4
BER		<1E-11	<6.9E-12	<1E-12	<1E-12
Energy efficiency <pJ/b>	Laser	**0.76**[d]	-	-	-
	Thermal	**0.57**[e]|0.49[f]	<0.6	-	-
	TRX + Clock + TT circuits	**1.26**	4.9-5	3.8 (RX only)	0.96 (RX only)
	Total	**2.59**|2.51	-	-	-

[a] **Bi-directional, 1-PHY deep** (0.6mm). [b] Bi-directional. [c] Measured with one data lane and one FWDCLK ON, PRBS7 pattern. [d] Assuming **10%** fiber-coupled laser WPE. [e] Measured with the bench laser. [f] Estimated based on the measured heater efficiency, covering 3σ of ring and laser variation.

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Figure IA24-O. NVIDIA ISSCC 2026 comparison table reproduced in the owner-supplied May 31 article, body image 18. Research link metrics with original footnotes; not a qualified memory system or commercial unit-cost quotation.

Original table metric	Value / boundary	Model treatment
Data throughput per fiber	256 Gb/s	Link demonstrator; no company shipment assumption
EIC shoreline density	0.8 Tb/s/mm bidirectional	0.4 in one direction; do not compare to an unlike-direction figure
Total energy	2.59 / 2.51 pJ/bit	Includes modeled laser/thermal terms; not whole-system memory energy
Laser efficiency assumption	10% fiber-coupled wall-plug efficiency	An assumption underlying the energy calculation
Receiver sensitivity condition	One data lane plus one clock lane; PRBS7	Does not establish every-lane, every-workload memory performance

Architecture ledger. The subjective probability of at least one material optical-attached DRAM program by end-2029 is **25%**. The link result and credible engineering path justify nonzero weight; a qualified memory platform's cohort and economics remain distinct. The conditional bridge is displaced HBM sales, plus recaptured LPDDR/other DRAM sales, less avoided HBM wafer/stacking cost, plus replacement DRAM/interface cost and incremental R&D/qualification expense. A bounded proxy was attempted by exposing 0-100% of the company's HBM pool; that range cannot be narrowed to an identifiable accelerator cohort, attachment rate or deployment schedule, and the replacement revenue/cost ratio can change even the sign of the earnings effect. The broad envelope overlaps heterogeneous-memory growth already in the reference model. **The incremental revenue and operating-profit contribution is zero in every current quarter for that specific non-identification and overlap reason**, not because official confirmation is required. The earliest author timing can overlap 2028; this is not dismissed merely as outside the valuation window.

The separate subjective probability of the author's **90% peak-volume HBM decline by 2033-2036 is 20%**. Its conditional endpoint is 10% of a future HBM peak, whose volume unit is not defined by the source. It is outside the forecast ending in 2030 and adds **zero** to the current earnings path. It is the same substitution mechanism as the 25% program hypothesis, so the two probabilities are not multiplied or applied as independent haircuts. This is a watch hypothesis, not a terminal-value or multiple adjustment.

Raise the program weight toward 40% when a named architecture provides a defensible affected cohort and comparable memory latency, delivered bandwidth, utilization and total cost; reduce it toward 10% after missed design/qualification milestones. Credible supplier/channel evidence can support that bridge before official commercialization. Revisit the distant 20% outcome only when multiple large architectures show net HBM substitution, including the HBM retained as cache. Mere link demonstrations and reposts do not independently increase either weight.

For Samsung, the replacement bridge must run through the separate HBM and conventional-DRAM vectors: subtract only displaced HBM sales, add only Samsung-captured replacement bits, and reconcile wafer, packaging, foundry and qualification costs on a consolidated basis. It cannot treat freed wafers as immediate saleable bits or stack a new LPDDR market on the existing DRAM residual. With those cohort variables unresolved, the article changes the architecture risks and tests, while the active quarterly operating path, 6× OP method, analytical Base 83 and current Excel Base 83 remain unchanged. The old 99 override was removed October 7.

2. Earnings, guidance and source evidence

October 8 issuer preliminary guidance, incorporated October 11

The issuer guides Q3 2026 consolidated sales of **195T** (range **194-196T**) and operating profit of **107.4T** (range **107.3-107.5T**) under K-IFRS. The headline values are the medians. The resulting 107.4 / 195 = **55.076923076923%** operating margin is our arithmetic on preliminary guidance. Samsung supplies this before external audit is complete; the full earnings call is scheduled for **October 29, 2026, 10:00 KST**. Do not label these figures final actual results. [SAMS-Q3G, SAMS-Q3IR, SAMS-Q3CALL]

Q3 2026 consolidated bridge	Prior model	Preliminary guidance	Unallocated difference
Revenue, KRW T	187.586027357573	195.000000000000	+7.413972642427
Operating profit, KRW T	103.138027357573	107.400000000000	+4.261972642427
Operating costs, KRW T	84.448000000000	87.600000000000	+3.152000000000

The current release does not provide a product or segment bridge. HBM, conventional DRAM, NAND and non-memory forecasts therefore remain explicitly provisional estimates. A separate unallocated revenue residual of **7.413972642427T** and OP residual of **4.261972642427T** reconcile those estimates to the issuer totals. Neither residual is assigned to Memory, product pricing, bit volume or yield. This is a genuine preliminary earnings event, rather than a new earnings period inferred from a digest title. TrendForce's October 7 evening EDT relays correspond to October 8 morning KST and refer to this same release.

Rule 16b persistence diagnostic, not an additional active forecast. Assign a **25%** subjective probability that part of the unexplained Q3 beat persists for the next two quarters. The conditional range is **0-100%** of the residual per quarter: revenue 0-7.413972642427T, operating cost 0-3.152T and OP 0-4.261972642427T, assuming the same residual mix for this bounded stress test only. The probability-weighted pre-overlap envelope is revenue 0-1.853493160607T, cost 0-0.788T and OP 0-1.065493160607T per quarter; over two quarters OP is 0-2.130986321214T. One persistence outcome covers both quarters; it is not two independent shocks.

Apply a **100% overlap adjustment** to this diagnostic when reconciling it to the active path: Q4 2026 already models **121.08894968382404T OP**, 13.68894968382404T above Q3 guidance, and Q1 2027 already models **144.19721000342682T**. Existing NAND and HBM assumptions already produce substantial further profit growth. The single consolidated beat does not identify additional product pricing, saleable-bit output or a persistent cost reduction beyond that path. The **active incremental revenue, cost and OP contribution from this persistence diagnostic is zero**. That zero follows overlap and lack of a separable incremental driver after the bounded proxy, not a requirement for official confirmation. Stronger product/mix/cost evidence can change the active model before a final release. Reassess at the October 29 segment disclosure or earlier credible product-specific evidence.

Samsung's July 30 issuer release reports Q2 2026 consolidated revenue of approximately KRW 171.5T and operating profit 89.5T. The official deck separately reports **Memory revenue 120.8T**, DS revenue 127.5T and **DS operating profit 89.2T**. The 89.2T is not standalone Memory profit. DX revenue 48.0T, Display 7.5T and Harman 4.6T require approximately 16.1T of eliminations when combined with DS. The model's non-memory net-sales allocations remain proxies. [SAMS-1, SAMS-2]

The latest company deck describes HBM4 sales growth, HBM4E sampling and server-centered memory demand with continuing capacity constraints. The prior call record gives Q3 DRAM bit growth in the mid-single digits and NAND in the high-single digits. Multi-year agreements and product price floors improve visibility but do not prove an upper price collar or guaranteed future margin for every product. Those July comments support the inherited operating path; the October 8 preliminary earnings release is treated separately above. [SAMS-2, SAMS-3]

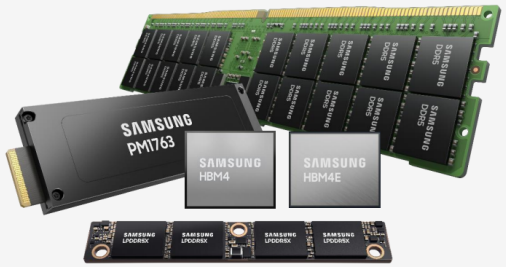
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Memory

Results

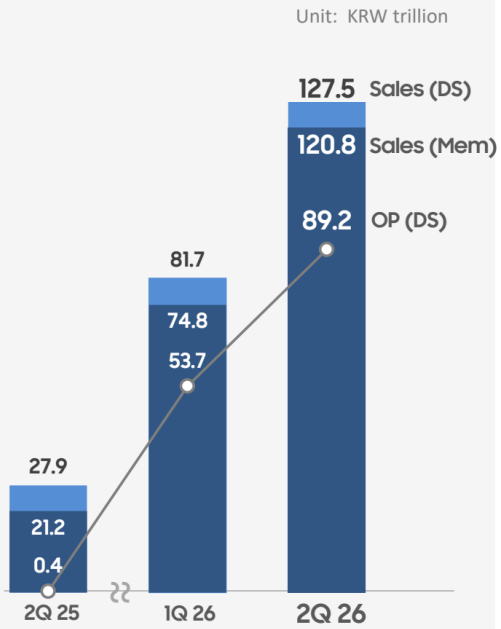
2Q 2026 Results

- **Achieved another record-breaking quarterly earnings**
  - Proactively addressed AI demand with primary focus on server despite limited capacity
  - Continued industry-wide upward price trend
- **Solidified market competitiveness via industry-leading products**
  - Achieved record-high server revenue mix
  - Scaled up HBM4 sales with industry-leading performance
  - Shipped industry's 1st HBM4E samples to major customers



※ Image : Samsung HBM4/E, DDR5 RDIMM, SOCAMM2, V9 eSSD

DS Results



Outlook

H2 2026

- **Expect robust demand centered on server backed by continued AI Infrastructure CapEx**
  - Strong server demand driven by broader adoption of agentic AI
  - Accelerating growth in demand for server DRAM, Enterprise SSD and HBM expected to keep the market undersupplied despite partial demand moderation in mobile and PC
- **Continue leading the market via technology leadership**
  - Seek to optimize portfolio based on demand trends by application and customer feedback; supply constraints to continue in spite of efforts to increase production
  - Plan to lead the market by addressing strong demand with a focus on high-value-added products, such as HBM4, DDR5, SOCAMM2, etc.
  - Extend the leadership into next-generation AI platforms, following the industry's 1st HBM4E sample shipment as well as the Gen6 and UFS 5.0 market penetration in H1.

7

Figure 1. Samsung Q2 2026 earnings presentation, page 7, July 30. Memory sales are reported; HBM qualification and demand commentary support the thesis. DS operating profit must not be misidentified as Memory profit. Source SAMS-2.

New HBM price, mix and supply evidence

The owner-supplied KB Securities page attributes a rising HBM wafer allocation to Samsung's Korea Premium Week presentation. It projects conventional DRAM's share at 73% in 2025,65% in 2026 and 59% in 2027. KB expects HBM4 to represent 80% of Samsung HBM revenue in 2027 versus 40% in 2026, and infers more than 100% HBM ASP growth. These mix and price forecasts are analyst expectations, not completed company sales. The original image is preserved here at the point of use. [KB-1]

FIRST TO THE MARKET

October 1, 2026

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Prolongation of DRAM Supply Shortage

Commodity DRAM shortage not surprising

— At Korea Premium Week 2026, SEC projected that HBM will account for more than a third of global DRAM wafer production capacity by 2027. With a new product category emerging and consuming 30% of total wafer capacity in just 2-3 years, the company noted that a commodity DRAM shortage is unsurprising under the current supply structure. Accordingly, the share of commodity DRAM (minus HBM) is expected to gradually decline from 73% in 2025 to 65% in 2026E and 59% in 2027E.

HBM ASP to rise more than 100% in 2027E

— Even if the three DRAM makers (incl. SEC) allocate 30% of total DRAM wafer capacity to HBM by end-2027, HBM is expected to account for only 13% of total supplied DRAM storage capacity as HBM requires more than 3x as much wafer input as commodity DRAM. This means that expanding HBM output substantially erodes commodity DRAM output capacity. HBM4 is expected to account for 80% of SEC's 2027E HBM revenue (vs. 40% in 2026E), leading us to believe that HBM ASP will rise more than 100% YoY in 2027.

Expectations for share retirement and higher dividends

— Assuming SEC has KRW110tn remaining for its 2024-26 shareholder return program, we expect the company to allocate KRW70tn to cash dividends (KRW30tn in 3Q26E, KRW40tn in 4Q26E), and KRW40tn to share buyback/retirement. Given the criteria for separate taxation of dividend income and the 25% payout ratio requirement, we believe the company will be able to increase its dividend payout to KRW40tn in 4Q26E after paying KRW30tn in 3Q26E, leaving enough funding to buy back and retire shares. We expect SEC to average quarterly OP of more than KRW100tn in 2H26E, and forecast 2026E OP of KRW368tn and 2027E OP of KRW555tn. Assuming the next 3y program also commits 50% of FCF to shareholder return, we expect total 3y shareholder returns to increase 4x from KRW140tn under the current 3y program to KRW600tn under the next.

Figure 2. Owner-supplied KB Securities note dated October 1, 2026, Jeff Kim and Changmin Lee. Company-attributed wafer commentary and analyst HBM/earnings/capital-return forecasts have different evidence status. The full exhibit is retained; its analyst valuation and buyback expectations are not adopted.

TrendForce's September 29 original research forecasts **121% growth in blended HBM ASP in 2027**, reflecting constrained output and HBM4/HBM4E mix. Buyers may reduce stack capacity from 12-Hi to 8-Hi, which lowers capacity per stack while increasing base-die cost per bit. The headline ASP denominator is not defined clearly enough to assert it is a pure like-for-like per-bit price. Our use of the forecast at fixed saleable-bit output is an explicit normalization assumption, and contributes to the 50% realization weight. Neither a separate HBM4 premium nor an 8-Hi premium is added. [TF-HBM]

The wafer figures are directional supply evidence, not a Samsung revenue split. A fall from 65% to 59% is a 9.23% relative decline in allocation only if total wafer capacity is constant. About 10.17% growth in total capacity offsets that allocation decline before productivity and yield changes. The same page's 30%-wafer/13%-bit illustration is not combined mechanically with 59% commodity share as one exact production schedule.

KB projects 2027 company OP 555T, below our inherited 626.710T. This is a counterweight to indiscriminate upgrades, not a replacement for the existing model. Its capital-return forecasts do not specify a reliable company-authorized cancellation schedule for our share denominator; future cash distributions cannot be added both to the profit valuation and as an independent profit layer.

3. Analytical score

The category arithmetic gives **50/60 = 83/100** after the existing -4-point risk deduction. The portfolio Excel Base remains **83/100**, matching the October 7 owner override ledger. Correcting the prior artifact's stale 99 text is not a new score change. Risk-class detail appears in section 7.

Category	Points	Maximum	Basis
Core thesis	17	20	Memory oligopoly and process breadth, assessed at the listed conglomerate level.
Cycle stage	10	10	Current memory allocation and HBM4 ramp.
TAM expansion	10	10	DRAM, HBM, NAND, custom logic and physical AI extend the addressable markets.
Catalysts	5	5	HBM4/HBM4E, advanced foundry nodes and capital-return decisions.
Macro	10	10	AI infrastructure demand and limited qualified capacity.
Alpha	2	5	Local-language contract and share-class work adds detail; broad company coverage limits asymmetry.
Risks	-4	0	Four classes: competitive supply/pricing; HBM/foundry execution; conglomerate governance/capital allocation; customer/device demand and mix.

4. Product decomposition, probability and cost bridge

Reconstruct the embedded HBM exposure before repricing

Mirae's May 7 research forecasts 2026 HBM revenue 33.8T. Its September 7 update forecasts 2027 HBM bit growth 57% and ASP growth 49%. These are named-source estimates used to reconstruct an HBM allocation **within** our existing Memory forecast. The old artifact did not explicitly disclose a 49% HBM price assumption; this reconstruction is a reasoned proxy, not proof of its historical hidden inputs. [MA-1, MA-2]

Reference 2027 HBM revenue =33.8 x 1.57 x 1.49 =**79.06834T**. Allocate it across 2027 in proportion to the existing quarterly non-NAND memory revenues, totaling 481.802951T. HBM's reference share is 16.41093% of that pool; conventional DRAM is the residual. For 2026 forward rows, the allocation proxy is 33.8 / (246.9 x 1.443) =9.48701%, using Mirae's 2026 DRAM USD forecast and annual FX assumption. This is a calendar-average allocation, not a claim that each 2026 quarter has identical product mix. Later reference HBM revenue scales with the retained non-NAND curve at the 2027 allocation. All reference product totals reproduce the original Memory revenue before any new increment.

As a scale cross-check, Mirae's September 2026 DRAM revenue growth and separate HBM/conventional bit-and-price growth imply roughly 34.14T of 2026 HBM revenue, close to the 33.8T anchor. Rounding and FX prevent precision claims. HBM remains a minority of Samsung's memory economics despite its rapid growth. No extra fab output is inserted into the old consolidated trajectory.

Calibration input	Value	Evidence / status
2026 HBM revenue	33.8T	Mirae May 7 forecast; stale but cross-checked scale
Reference 2027 HBM bit growth	57%	Mirae Sep 7 forecast; not official output
Reference 2027 HBM ASP growth	49%	Mirae Sep 7; reconstructed overlap proxy
Conditional 2027 blended ASP growth	121%	TrendForce industry forecast applied as Samsung proxy
Realization weight	50%	Subjective outcome probability
2027 active blended ASP growth	85%	49% +50% x(121%-49%)
2027 conditional HBM revenue	117.27586T	33.8 x 1.57 x 2.21
2027 active HBM revenue	98.17210T	79.06834 +50% x 38.20752



HBM price/mix: count the incremental change once. Model assumptions, not company guidance. Numerical table follows.

Index,2026=100	2026	2027
Reference	100	149
Active weighted	100	185
Conditional endpoint	100	221

Explicit probability ledger

HBM-01: 50% probability that Samsung realizes the conditional 121% 2027 blended pricing/mix increase on the calibrated output. TrendForce has the owner's top-tier weighting; KB independently publishes a directionally similar greater-than 100% Samsung estimate, and company ramp evidence supports the product transition. Exact company contracts, headline price units and qualification/stack mix remain uncertain; researcher inputs may share underlying industry checks. The weight therefore applies to the economic outcome, not source authenticity, and is not a claim of statistically independent confirmation.

Active HBM revenue =reference HBM revenue +p x(conditional revenue-reference revenue). Active HBM cost =reference cost +p x(conditional cost-reference cost). Both outcomes use the same saleable output, so the calculation adds pricing/mix economics once, not another volume ramp. No increment is assigned to 2026; the 2027 price premium is retained through 2028, halved in 2029, and removed in 2030. The later fade is our uncertainty assumption, not a new annual price forecast. Costs fade in the same proportion.

DRAM-01: 60% shortage-persistence judgment, zero separate sales addition. The wafer reallocation signal supports the existing conventional DRAM path; adding another price multiplier from the same shortage would duplicate already modeled scarcity. We explicitly attempted a wafer-share-to-output bridge, but total productive wafers, yield, node productivity and Samsung capture are missing. Allocation alone cannot identify extra sales. This zero reflects overlap and the missing conversion, not a demand for official confirmation. The HBM branch still has an active numerical contribution.

Raise HBM-01 toward 65% if company-specific contract pricing and qualification shipments establish comparable-unit conversion; reduce it toward 25% if concessions, lower-capacity stacks or ramp misses dilute the forecast. Revisit the embedded 49% assumption if newer evidence shows more of this price gain was already in the inherited curve. Increase cost stress if base-die, packaging or yield execution disappoints. No independent Base-score increase is awarded for the same profit increment.

Production, packaging, yield and operating expenses

The existing Memory operating-cost envelope is retained as the starting point. Allocate it using revenue-weighted relative cost factors **HBM 2.0, conventional DRAM 1.0, NAND 1.25**, normalized each quarter to sum exactly to the prior cost. These factors are analytical allocation assumptions, reflecting HBM's additional processing and packaging burden; they are not reported product margins. Conventional DRAM's extreme retained scarcity margin remains visible. The split makes that risk inspectable rather than imposing an unrelated margin cap.

For HBM,80% of the reference operating cost is allocated to production/COGS and 20% to operating expenses. Production cost is split 65% front-end and 35% packaging. Under the conditional outcome, front-end cost per input rises 10% and packaging cost 20% above the embedded ramp, before a yield adjustment from 95% to 90%. These yields are stress proxies, not measured Samsung yields. Holding saleable output fixed means more input is required; the model does not also subtract units sold. COGS factor =[0.65 x 1.10 +0.35 x 1.20] x 0.95/0.90 =1.19806. Additional R&D/SG&A equals 5% of the incremental HBM sales. Existing reference opex stays in place. These assumptions represent mix/ramp cost stress accompanying the blended-ASP forecast, rather than suggesting a like-for-like price increase inherently consumes more wafers.

Conditional HBM COGS =reference HBM cost x 80% x[1 +fade x(1.19806-1)]. Conditional HBM opex =reference cost x 20% +5% x(conditional sales-reference sales). Apply the 50% weight to the resulting total cost difference. Manufacturing cost includes the operating burden of production; capex is not deducted again dollar-for-dollar from operating profit. Company-disclosed gross margins and product opex are unavailable, so these synthetic allocations must never be presented as reported financials.

Cost assumption	Reference	Conditional
Relative cost weights HBM / DRAM / NAND	2.0 /1.0 /1.25	Reference split preserved
HBM COGS / opex split	80% /20%	Extra opex separately below
Front-end / packaging share of COGS	65% /35%	Same split before cost stresses
Front-end cost per input	1.00	1.10
Packaging cost per input	1.00	1.20
Saleable-yield proxy	95%	90%
Additional R&D/SG&A	None	5% of added HBM sales
Outcome weight	50% reference	50% conditional

2027 HBM economic bridge	KRW T
Reference HBM revenue	79.068
Conditional revenue	117.276
Weighted revenue increment	19.104
Weighted cost increment	2.661
Weighted OP increment	16.443

Existing NAND assumptions

Active NAND pricing and transmission

The active NAND benchmark is **calendar Q3 2026 +23%, Q4 2026 +20% and Q1 2027 +15% sequentially**. Q3/Q4 retain the upper bounds of the owner-supplied TrendForce ranges, ranked Tier 1 alongside Serenity. The added Q1 forecast comes from the owner-supplied Jeff Pu/Jukan summary citing TrendForce. The owner authorized this combined case on September 21. The quoted analyst path was +21%/+17%/+15%; the active case retains the owner's higher 2026 selections. [TF-1, JF-1]

With Q2 2026 indexed to 1.000, active calendar price indexes are **1.230 in Q3, 1.476 in Q4 and 1.6974 in Q1 2027**. The Q1 level is 69.74% above Q2 2026. The +15% is applied once in Q1; it does not recur every quarter. Later forecasts retain their existing operating trajectory at this higher benchmark level. Persistence through 2030 is a modeling assumption, not a forecast supplied by TrendForce or the quoted analysts.

The inherited company curves did not contain an auditable NAND ASP/bit split. The current calculation therefore uses **reference-price calibration divisors of 1.150 for Q3 and 1.2075 for Q4 onward**. These reproduce the supplied curve at the previous upper-bound benchmark and are calculation inputs, not a second active price

forecast. Reference-price NAND revenue = calibration NAND revenue / reference index. Active NAND revenue = reference-price NAND revenue x active index. The underlying volume/mix proxy is held constant during this repricing; it is not a measured bit shipment series.

This full-transmission operating case replaces the price component rather than applying another 23% and 20% on top of already priced revenue. This NAND bridge adds no DRAM/HBM price uplift; the separate HBM-01 branch above provides the new HBM increment. No new fab capacity or extra buyback is assumed. Company contracts, qualification, mix and timing remain the tests of realized earnings.

Samsung NAND allocation and consolidation

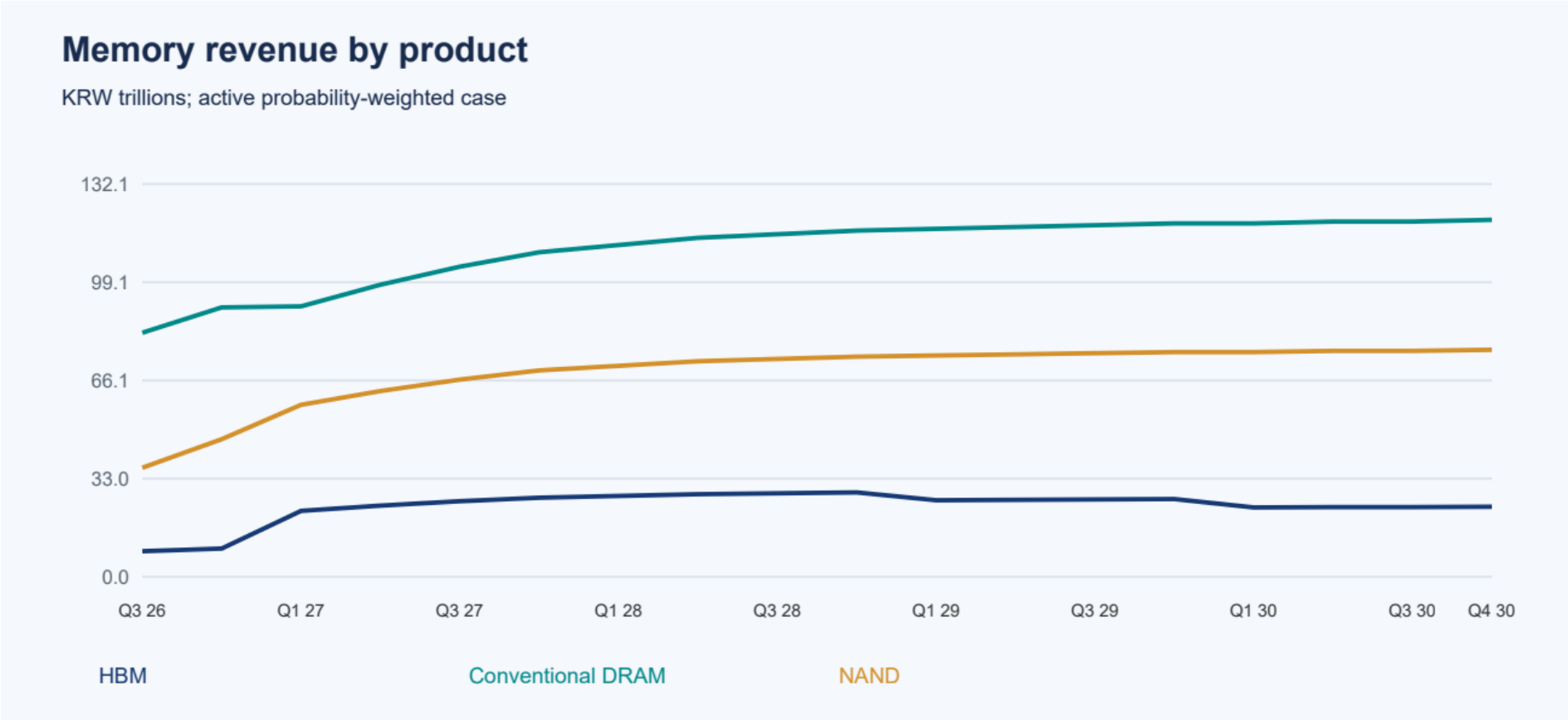
TrendForce's matched Q2 estimates imply NAND / (DRAM + NAND) = **23.06 / (60.98 + 23.06) = 27.4393%**. This ratio allocates the modeled Memory revenue pool; it is not a Samsung-disclosed segment percentage or a new consolidated sales line. Holding that allocation through the forecast is a LOW-confidence modeling assumption. HBM remains within the non-NAND DRAM residual and receives no NAND price uplift. [TF-2, TF-3]

For each quarter, calibration NAND revenue equals the reference Memory curve x 27.4393%. After applying the active price index, Memory revenue equals non-NAND Memory plus active NAND. Reference Memory operating cost, including production expense and divisional opex, is retained in absolute currency from the supplied curve before the separate HBM cost increment. Consequently the incremental price revenue flows one-for-one into Memory OP in this operating ceiling case. Memory OPM is derived from revenue and cost; the HBM branch changes both. NAND and conventional DRAM allocated costs receive no new pricing overlay.

The allocation is applied to the existing consolidated-model Memory pool. No internal sale is added again and DX, foundry and other net-sales proxies remain separate. Existing device-cost pressure stays in the DX path. Any additional demand loss, external NAND purchasing cost, employee incentive expense or unrecovered device margin pressure would reduce group profit relative to this case and is not quantified here. Product scope and internal-elimination differences between industry estimates and company accounts remain an allocation risk.

5. Quarterly operating and valuation model

Calendar quarters are used. Q3 2026 consolidated revenue and OP are issuer preliminary guidance; all Q3 product/segment splits remain LOW-confidence model estimates with a separate unallocated reconciliation. Q4 2026 onward and every product schedule remain modeled. No source provides this full quarterly schedule. Display rounding does not alter exact calculations.



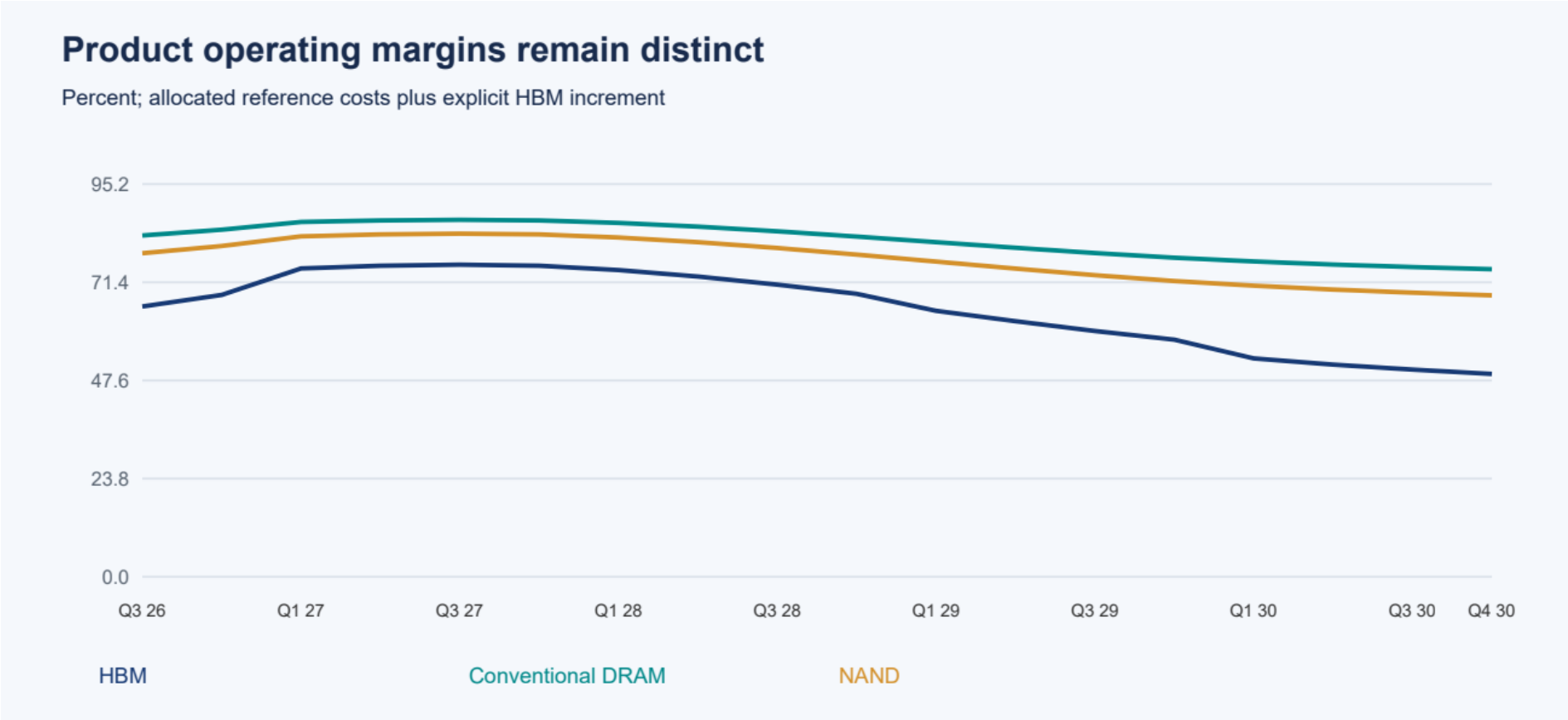
Memory revenue by product. Model assumptions, not company guidance. Numerical table follows.

Quarter	HBM rev	Conv DRAM rev	NAND rev	Memory total
Q3 2026	8.605	82.096	36.685	127.386
Q4 2026	9.500	90.634	46.286	146.420
Q1 2027	22.177	90.979	57.858	171.014
Q2 2027	23.952	98.258	62.486	184.696
Q3 2027	25.430	104.323	66.344	196.097
Q4 2027	26.613	109.175	69.429	205.217
Q1 2028	27.204	111.601	70.972	209.778

Quarter	HBM rev	Conv DRAM rev	NAND rev	Memory total
Q2 2028	27.796	114.027	72.515	214.338
Q3 2028	28.091	115.240	73.286	216.618
Q4 2028	28.387	116.453	74.058	218.898
Q1 2029	25.759	117.060	74.444	217.262
Q2 2029	25.892	117.666	74.829	218.388
Q3 2029	26.026	118.273	75.215	219.514
Q4 2029	26.159	118.879	75.601	220.639
Q1 2030	23.339	118.879	75.601	217.820
Q2 2030	23.459	119.486	75.987	218.931
Q3 2030	23.459	119.486	75.987	218.931
Q4 2030	23.578	120.093	76.372	220.042

Product operating costs and profit

Quarter	HBM op cost	Conv DRAM cost	NAND cost	HBM OP	Conv DRAM OP	NAND OP
Q3 2026	2.964	14.139	7.897	5.641	67.957	28.788
Q4 2026	3.006	14.338	9.153	6.494	76.296	37.133
Q1 2027	5.591	12.685	10.084	16.586	78.294	47.774
Q2 2027	5.882	13.330	10.596	18.070	84.928	51.890
Q3 2027	6.178	13.995	11.125	19.252	90.328	55.218
Q4 2027	6.535	14.811	11.774	20.078	94.364	57.656
Q1 2028	6.966	15.813	12.570	20.239	95.788	58.402
Q2 2028	7.554	17.188	13.663	20.242	96.839	58.852
Q3 2028	8.187	18.674	14.844	19.905	96.567	58.442
Q4 2028	8.905	20.362	16.187	19.482	96.091	57.871
Q1 2029	9.142	22.056	17.533	16.616	95.004	56.910
Q2 2029	9.841	23.767	18.893	16.051	93.900	55.936
Q3 2029	10.511	25.405	20.195	15.515	92.868	55.020
Q4 2029	11.113	26.879	21.367	15.046	92.001	54.234
Q1 2030	10.976	27.954	22.222	12.363	90.925	53.379
Q2 2030	11.386	28.997	23.051	12.073	90.489	52.936
Q3 2030	11.669	29.718	23.623	11.790	89.768	52.363
Q4 2030	11.977	30.502	24.247	11.601	89.590	52.125



Product operating margins remain distinct. Model assumptions, not company guidance. Numerical table follows.

Quarter	HBM OPM	Conv DRAM OPM	NAND OPM	Memory OPM
Q3 2026	65.56%	82.78%	78.47%	80.37%
Q4 2026	68.36%	84.18%	80.23%	81.90%
Q1 2027	74.79%	86.06%	82.57%	83.42%
Q2 2027	75.44%	86.43%	83.04%	83.86%
Q3 2027	75.71%	86.58%	83.23%	84.04%
Q4 2027	75.44%	86.43%	83.04%	83.86%
Q1 2028	74.40%	85.83%	82.29%	83.15%

Quarter	HBM OPM	Conv DRAM OPM	NAND OPM	Memory OPM
Q2 2028	72.82%	84.93%	81.16%	82.08%
Q3 2028	70.86%	83.80%	79.75%	80.75%
Q4 2028	68.63%	82.51%	78.14%	79.24%
Q1 2029	64.51%	81.16%	76.45%	77.57%
Q2 2029	61.99%	79.80%	74.75%	75.96%
Q3 2029	59.61%	78.52%	73.15%	74.44%
Q4 2029	57.52%	77.39%	71.74%	73.10%
Q1 2030	52.97%	76.49%	70.61%	71.93%
Q2 2030	51.46%	75.73%	69.66%	71.03%
Q3 2030	50.26%	75.13%	68.91%	70.31%
Q4 2030	49.20%	74.60%	68.25%	69.68%

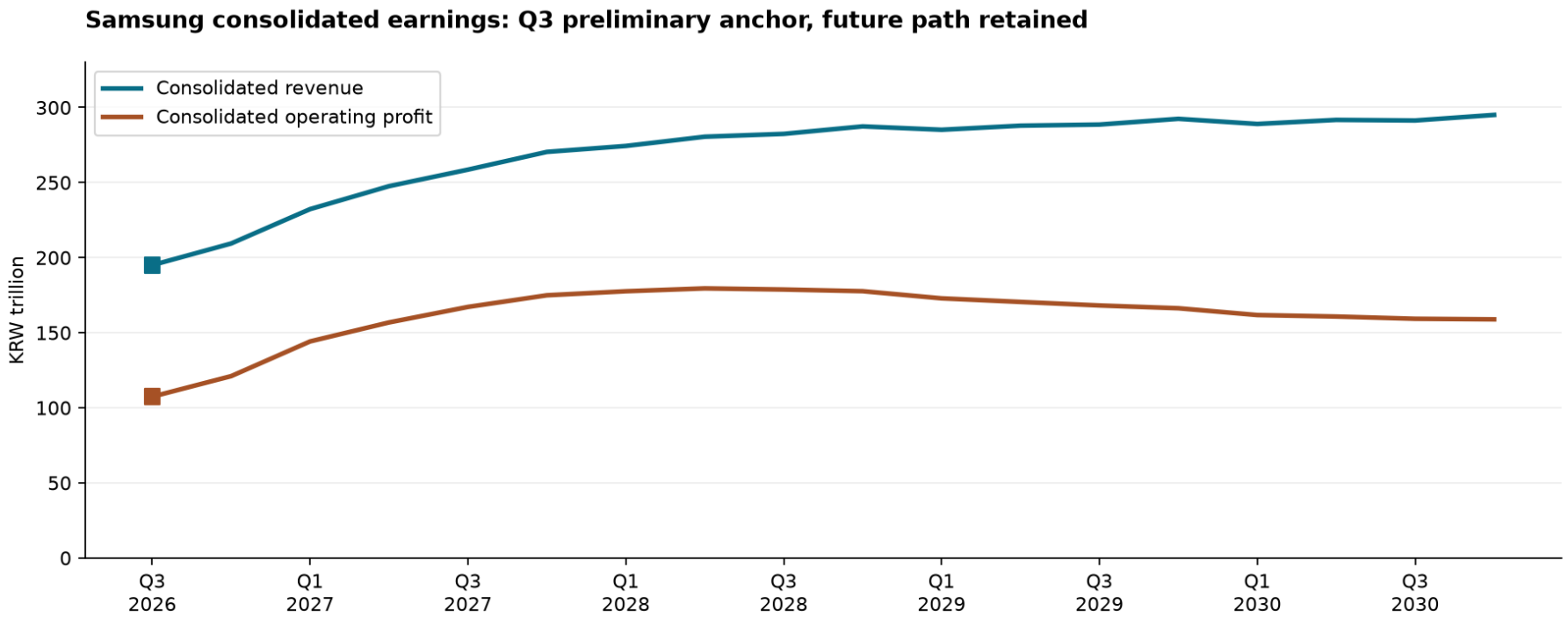
Incremental quarterly bridge

Quarter	Ref HBM rev	Conditional rev	Weighted rev +	Weighted cost +	Weighted OP +
Q3 2026	8.605	8.605	0.000	0.000	0.000
Q4 2026	9.500	9.500	0.000	0.000	0.000
Q1 2027	17.862	26.493	4.316	0.610	3.705
Q2 2027	19.291	28.612	4.661	0.648	4.013
Q3 2027	20.482	30.379	4.949	0.683	4.266
Q4 2027	21.434	31.792	5.179	0.720	4.459
Q1 2028	21.911	32.498	5.294	0.757	4.537
Q2 2028	22.387	33.205	5.409	0.805	4.604
Q3 2028	22.625	33.558	5.466	0.854	4.612
Q4 2028	22.863	33.911	5.524	0.910	4.614
Q1 2029	22.982	28.535	2.776	0.482	2.295
Q2 2029	23.101	28.683	2.791	0.509	2.282
Q3 2029	23.220	28.831	2.805	0.535	2.270
Q4 2029	23.339	28.979	2.820	0.559	2.260
Q1 2030	23.339	23.339	0.000	0.000	0.000
Q2 2030	23.459	23.459	0.000	0.000	0.000
Q3 2030	23.459	23.459	0.000	0.000	-0.000
Q4 2030	23.578	23.578	0.000	0.000	0.000

HBM reference and conditional values are diagnostic components of one weighted operating case. For Q4 2026 onward the prior total-company reference path is recovered when HBM-01 probability is zero; Q3 2026 consolidated preliminary guidance remains fixed and is reconciled separately. No double multiplication by the analytical score or by a final valuation probability occurs.

Retained non-memory businesses

Quarter	Foundry/LSI rev	OPM	DX net rev	OPM	Other net rev	OPM
Q3 2026	16.000	2.00%	38.000	-1.80%	6.200	18.00%
Q4 2026	16.500	2.80%	40.000	-1.20%	6.500	18.20%
Q1 2027	17.000	3.60%	38.000	-0.60%	6.300	18.40%
Q2 2027	17.500	4.40%	39.000	0.00%	6.400	18.60%
Q3 2027	18.000	5.20%	38.000	0.60%	6.500	18.80%
Q4 2027	18.500	5.80%	40.000	1.20%	6.700	19.00%
Q1 2028	19.000	6.40%	39.000	1.80%	6.600	19.10%
Q2 2028	19.500	6.80%	40.000	2.40%	6.700	19.20%
Q3 2028	20.000	7.20%	39.000	2.80%	6.800	19.30%
Q4 2028	20.500	7.50%	41.000	3.20%	7.000	19.40%
Q1 2029	21.000	7.70%	40.000	3.50%	6.900	19.40%
Q2 2029	21.500	7.90%	41.000	3.80%	7.000	19.50%
Q3 2029	22.000	8.00%	40.000	4.00%	7.100	19.50%
Q4 2029	22.500	8.10%	42.000	4.20%	7.300	19.60%
Q1 2030	23.000	8.20%	41.000	4.40%	7.200	19.60%
Q2 2030	23.500	8.30%	42.000	4.50%	7.300	19.70%
Q3 2030	24.000	8.40%	41.000	4.60%	7.400	19.70%
Q4 2030	24.500	8.50%	43.000	4.70%	7.500	19.80%



Squares: issuer preliminary guidance for Q3 2026. Lines after Q3: model estimates. Segment residuals remain unallocated.

Consolidated revenue and operating profit. Q3 2026 uses issuer preliminary guidance, not final actuals; Q4 2026 onward remains modeled. Numerical table and unallocated residual follow.

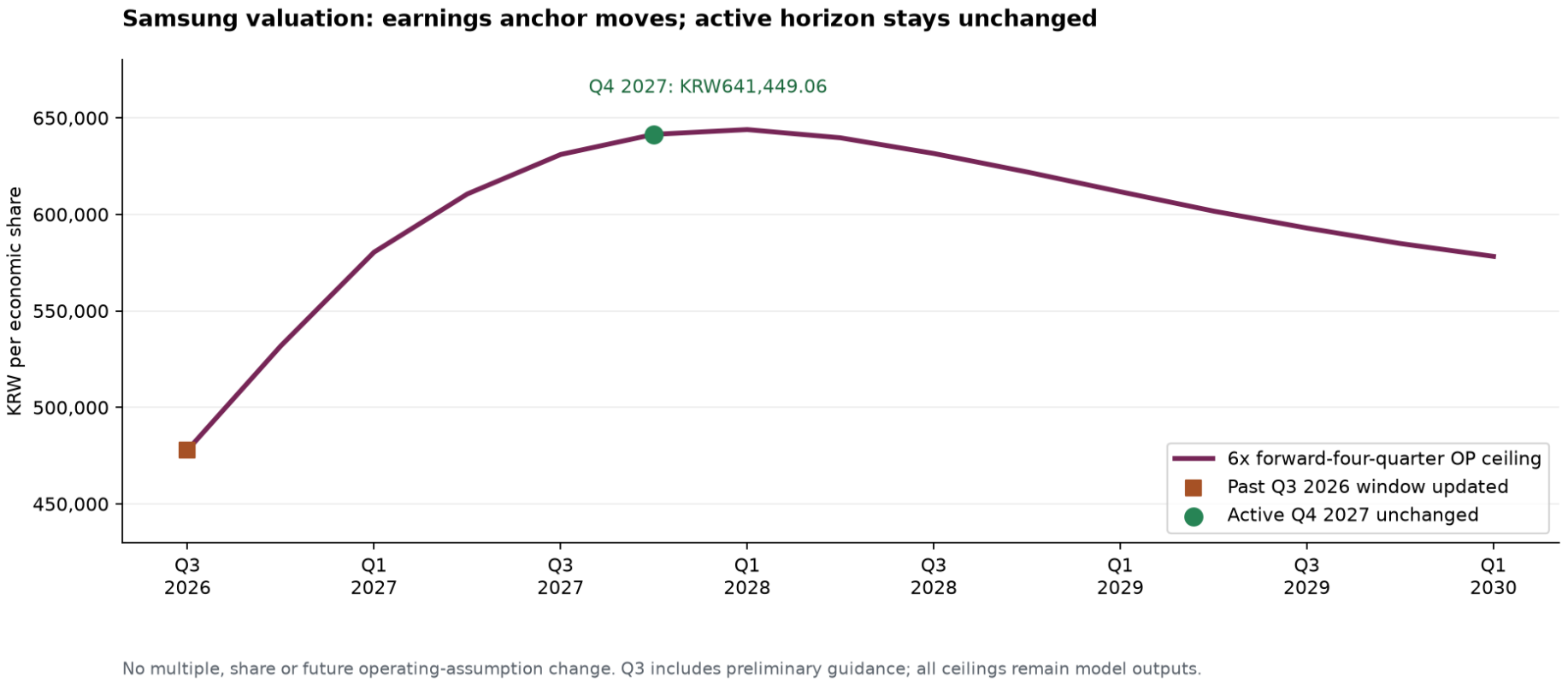
Quarter	Revenue	Memory OP est.	Non-memory OP est.	Unallocated OP	Group OP	Group OPM
Q3 2026 prelim.	195.000	102.386	0.752	4.262	107.400	55.08%
Q4 2026	209.420	119.924	1.165	0.000	121.089	57.82%
Q1 2027	232.314	142.654	1.543	0.000	144.197	62.07%
Q2 2027	247.596	154.888	1.960	0.000	156.848	63.35%
Q3 2027	258.597	164.798	2.386	0.000	167.184	64.65%
Q4 2027	270.417	172.098	2.826	0.000	174.924	64.69%

Quarter	Revenue	Memory OP est.	Non-memory OP est.	Unallocated OP	Group OP	Group OPM
Q1 2028	274.378	174.429	3.179	0.000	177.608	64.73%
Q2 2028	280.538	175.933	3.572	0.000	179.505	63.99%
Q3 2028	282.418	174.914	3.844	0.000	178.758	63.30%
Q4 2028	287.398	173.445	4.208	0.000	177.652	61.81%
Q1 2029	285.162	168.530	4.356	0.000	172.886	60.63%
Q2 2029	287.888	165.887	4.621	0.000	170.508	59.23%
Q3 2029	288.614	163.403	4.744	0.000	168.148	58.26%
Q4 2029	292.439	161.280	5.017	0.000	166.298	56.87%
Q1 2030	289.020	156.668	5.101	0.000	161.769	55.97%
Q2 2030	291.731	155.497	5.279	0.000	160.776	55.11%
Q3 2030	291.331	153.921	5.360	0.000	159.281	54.67%
Q4 2030	295.042	153.316	5.589	0.000	158.905	53.86%

Q3 2026 revenue includes the separate 7.413972642427T unallocated revenue reconciliation. Memory and non-memory OP remain segment estimates; adding the 4.261972642427T unallocated OP reconciles to preliminary group guidance. This residual does not represent a new reporting segment. Future unallocated increments are zero.

Calendar year	Revenue	Operating profit	OPM
2027	1,008.924	643.153	63.75%
2028	1,124.732	713.524	63.44%
2029	1,154.103	677.839	58.73%
2030	1,167.124	640.730	54.90%

Forward-four-quarter ceiling at unchanged 6x OP



Samsung quarterly valuation. Only the past Q3 2026 window changes; green marks the unchanged active Q4 2027 horizon. Valuation retains 6x OP and 6.648649811B shares.

The last column isolates the existing HBM pricing/cost branch versus its no-increment reference at the same quarter. For Q3 2026 both sides include the same new preliminary-guidance reconciliation; the residual is not attributed to HBM. This diagnostic is separate from the October 11 earnings-anchor revision.

Ceiling quarter	Exact earnings window	F4Q OP T	Multiple	Shares B	KRW/share	Existing HBM branch effect
Q3 2026	Q3 2026 to Q2 2027	529.534	6x	6.648649811	477,872.43	1.48%
Q4 2026	Q4 2026 to Q3 2027	589.318	6x	6.648649811	531,823.58	2.08%
Q1 2027	Q1 2027 to Q4 2027	643.153	6x	6.648649811	580,406.07	2.62%
Q2 2027	Q2 2027 to Q1 2028	676.563	6x	6.648649811	610,556.97	2.62%

Ceiling quarter	Exact earnings window	F4Q OP T	Multiple	Shares B	KRW/share	Existing HBM branch effect
Q3 2027	Q3 2027 to Q2 2028	699.220	6x	6.648649811	631,003.60	2.62%
Q4 2027	Q4 2027 to Q3 2028	710.795	6x	6.648649811	641,449.06	2.63%
Q1 2028	Q1 2028 to Q4 2028	713.524	6x	6.648649811	643,911.55	2.64%
Q2 2028	Q2 2028 to Q1 2029	708.802	6x	6.648649811	639,650.41	2.33%
Q3 2028	Q3 2028 to Q2 2029	699.805	6x	6.648649811	631,531.02	2.01%
Q4 2028	Q4 2028 to Q3 2029	689.194	6x	6.648649811	621,955.46	1.69%
Q1 2029	Q1 2029 to Q4 2029	677.839	6x	6.648649811	611,708.47	1.36%
Q2 2029	Q2 2029 to Q1 2030	666.722	6x	6.648649811	601,676.12	1.03%
Q3 2029	Q3 2029 to Q2 2030	656.990	6x	6.648649811	592,893.15	0.69%
Q4 2029	Q4 2029 to Q3 2030	648.123	6x	6.648649811	584,891.40	0.35%
Q1 2030	Q1 2030 to Q4 2030	640.730	6x	6.648649811	578,219.99	-0.00%

Ceiling =sum of the specified four quarterly operating profits x 6 / 6.648649811B economic shares x 1,000. The conversion reflects KRW trillions divided by billions of shares. No net-debt adjustment, tax conversion, net-income multiple or horizon substitution is introduced. The final three operating quarters support earlier valuation windows.

- Q3 2026, past valuation window: (107.400000 + 121.088950 + 144.197210 + 156.848249) x 6 x 1,000 /6.648649811 = KRW 477,872.43. Exact F4Q OP is 529.53440857130891T; exact ceiling is 477872.4315080118542883503359 KRW from the preserved cached inputs. The active Q4 2027 horizon excludes Q3 2026 and is unchanged.
- Q4 2026: (121.088950 + 144.197210 + 156.848249 + 167.183719) x 6 x 1,000 /6.648649811 = KRW 531,823.58.
- Q1 2027: (144.197210 + 156.848249 + 167.183719 + 174.923610) x 6 x 1,000 /6.648649811 = KRW 580,406.07.
- Q2 2027: (156.848249 + 167.183719 + 174.923610 + 177.607670) x 6 x 1,000 /6.648649811 = KRW 610,556.97.
- Q3 2027: (167.183719 + 174.923610 + 177.607670 + 179.505326) x 6 x 1,000 /6.648649811 = KRW 631,003.60.
- Q4 2027: (174.923610 + 177.607670 + 179.505326 + 178.758426) x 6 x 1,000 /6.648649811 = KRW 641,449.06.
- Q1 2028: (177.607670 + 179.505326 + 178.758426 + 177.652313) x 6 x 1,000 /6.648649811 = KRW 643,911.55.
- Q2 2028: (179.505326 + 178.758426 + 177.652313 + 172.885869) x 6 x 1,000 /6.648649811 = KRW 639,650.41.
- Q3 2028: (178.758426 + 177.652313 + 172.885869 + 170.508155) x 6 x 1,000 /6.648649811 = KRW 631,531.02.
- Q4 2028: (177.652313 + 172.885869 + 170.508155 + 168.147672) x 6 x 1,000 /6.648649811 = KRW 621,955.46.
- Q1 2029: (172.885869 + 170.508155 + 168.147672 + 166.297534) x 6 x 1,000 /6.648649811 = KRW 611,708.47.

- Q2 2029: (170.508155 + 168.147672 + 166.297534 + 161.768943) x 6 x 1,000 /6.648649811 = KRW 601,676.12.
- Q3 2029: (168.147672 + 166.297534 + 161.768943 + 160.775668) x 6 x 1,000 /6.648649811 = KRW 592,893.15.
- Q4 2029: (166.297534 + 161.768943 + 160.775668 + 159.280868) x 6 x 1,000 /6.648649811 = KRW 584,891.40.
- Q1 2030: (161.768943 + 160.775668 + 159.280868 + 158.904893) x 6 x 1,000 /6.648649811 = KRW 578,219.99.

Sensitivity to uncertain embedded pricing

Reconstructed embedded 2027 ASP	Outcome weight	Q3 2027 ceiling KRW	Change vs original
49.00%	50%	631,003.60	2.62%
75.00%	50%	624,283.56	1.53%
100.00%	50%	617,829.77	0.48%

Each diagnostic changes the HBM reference allocation within the same original total Memory pool; conventional DRAM is the residual, and associated product-cost allocation is recalculated. It is not an alternative active case or an additional price forecast. More price growth already embedded leaves less incremental upside. At zero HBM outcome probability, the pre-HBM reference path is recovered for future operating quarters and their valuation windows. Q3 2026 consolidated guidance remains the updated anchor; its unallocated reconciliation is not an HBM probability branch.

HBM outcome probability	Q3 2027 ceiling KRW
0.00%	614,880.75
25.00%	622,942.17
50.00%	631,003.60
75.00%	639,065.02
100.00%	647,126.45

Share-class and treasury-stock bridge

Samsung reports 5,846,278,608 common and 802,371,203 preferred issued shares at June 30, totaling 6,648,649,811. Treasury common shares are 82,086,705 at that date. Preferred shareholders participate in earnings; consolidated operating profit cannot all be assigned to common shares without an allocation. [SAMS-4]

The review uses 6.648649811B forward economic shares, assuming the currently issued shares ultimately participate, including treasury stock used for compensation. This is an explicit forward proxy, not the latest reported diluted weighted average. A zero-voting-premium economic allocation is used; no market-price discount is applied to preferred stock. Excluding the Q2 treasury balance produces 6.566563106B and would raise the ceiling by about 1.25%; this quantifies the remaining treasury-use sensitivity.

The August 21 resolution authorizes approximately KRW 15T of purchases for employee stock compensation during August 24-November 21. It specifies holding shares for incentive programs, so it does not establish a permanent cancellation benefit. Any future cancellation estimate must be separately scoped and probability-weighted with amount, timing and repurchase cost; this HBM pricing update does not change the share schedule. [SAMS-5]

Q3 2027 share sensitivity	Economic shares B	Ceiling KRW
5% fewer	6.316217320	664,214.31
Active	6.648649811	631,003.60
5% more	6.981082302	600,955.81

6. Industry relationships and read-throughs

TAM expansion and cross-book

AI inference expands conventional DRAM, HBM and enterprise storage together. Custom base dies, packaging, device AI and robotics provide adjacent demand; they must not be counted both as internal input sales and external system sales. LPK's glass-processing opportunity is one possible equipment flow-through, not an assumed Samsung order.

Cross-watch SK hynix and Micron on HBM qualification, supply growth and conventional memory allocation; watch SanDisk/Kioxia for NAND demand and contract terms. Samsung's price-floor disclosure supports a more stable contracted business but does not independently prove a floor for every future quarterly gross margin.

Samsung memory demand can validate industry investment in packaging and tools, but it does not establish an LPKF order or another supplier's capture. Strong memory economics may pressure Dell, Longsys and Penguin through replacement costs. Compare physical bit output and product mix before transferring a revenue-growth inference.

The active NAND benchmark is shared with the SNDK and Kioxia updates, but issuer exposure, recognition dates and operating cost treatment differ. The same percentage increase in NAND prices does not imply the same percentage increase in consolidated earnings or per-share value.

7. Risks and falsification tests

The following ledger makes the existing aggregate risk deduction explicit. It does not introduce another deduction or change the analytical score.

Risk class	Existing deduction
Competitive supply, pricing and allocation	-1
HBM qualification, packaging yield and foundry execution	-1
Conglomerate governance and capital allocation	-1
Customer, device demand and consolidated mix	-1

Risk class	Existing deduction
Total	**-4**

Risks and triggers

1. Supply/pricing response: test the shortage-duration thesis against qualified new capacity and actual pricing, not factory announcements alone.
2. HBM/foundry execution: monitor HBM4 conversion, HBM4E qualification and Taylor production acceptance. Customer-specific process wins determine revenue timing.
3. Governance/capital allocation: distinguish compensation purchases from cancellation, dividends and investment. Track preferred/common claims and future grants.
4. Demand/mix: monitor premium device sales, component inflation and server-versus-consumer mix.

The next earnings update should replace the HBM allocation, price normalization and synthetic product costs with stronger product/bit/ASP evidence and reconcile the share bridge to the latest dilution note. The operating ceiling remains conditional on the inherited memory path.

Numbered monitoring tests

#	Checkpoint	Evidence supporting the operating case	Counter-test / failure signal
1	HBM progress	Track qualified shipments and yield to revise the existing probability and capture assumptions; credible pre-qualification signals remain modelable under Rule 16b.	Qualification delays or disappointing yield reduce expected capture and timing; they do not impose an official-confirmation gate on breadcrumbs.
2	Consolidation	Reconcile memory and other businesses to group revenue and OP.	Unexplained eliminations or duplicate internal sales invalidate the bridge.
3	Economic shares	Reconcile common, preferred, treasury and dilutive instruments to the latest notes.	A common-only denominator overattributes group economics.
4	Diversification	Track profit conversion in foundry, devices and the residual businesses.	Memory strength offset by non-memory losses weakens consolidated leverage.

NAND pricing tests

Checkpoint	Required observation	Failure signal
Realized ASP	Track calendar Q3 +23%, Q4 +20% and Q1 2027 +15% benchmark against company NAND mix and realized ASP.	Contract caps, timing or discounts cause weaker transmission.
Output and cost	Reconcile saleable bits, mix, costs and incentive expense to the fixed reference-price curve.	Price gains are absorbed by lower units or additional costs.
Persistence	Test whether Q1 price gains persist as assumed in later forecasts.	ASP cuts or new capacity erase the retained price premium.

Additional tests for this update: separate a true per-bit price gain from larger stacks or richer products; confirm saleable-bit output and factory yield without counting the same volume twice; test the 49% embedded-price proxy against the inherited curve; monitor process, packaging and incentive costs. The largest remaining uncertainty is still the inherited high memory profitability, not the arithmetic of the new increment. The near 12-16 months carry more useful forecasting weight than 2029-2030; the explicit premium fade reflects that deterioration in visibility.

8. Sources, evidence gaps and model snapshot

- **SAMS-Q3G** - October 8, 2026, Samsung issuer Q3 preliminary guidance. Consolidated K-IFRS revenue 195T / OP 107.4T; ranges 194-196T /107.3-107.5T. Guidance, not final actuals. Original HTML retained with SHA-256.
- **SAMS-Q3IR** - October 8, 2026, Samsung IR pre-earnings disclosure 84775. Confirms units and preliminary status before external audit completion.
- **SAMS-Q3CALL** - Retrieved October 11, 2026, Samsung IR events. Full Q3 call scheduled October 29 at 10:00 KST.

Sources below are those documented in the supplied earnings review. LOW-confidence product allocations, future margins, capacity bridges and incomplete management-call records remain unresolved where stated. The current revision applies Framework 8.0 probability and visual requirements. Inherited sources are distinguished from the new evidence; their original gaps are not represented as newly resolved.

- **SAMS-1** · 2026-07-30 · Korean company Q2 results. Actual operating results and outlook, original language.
- **SAMS-2** · 2026-07-30 · Company Q2 presentation. Divisional revenue includes internal sales; consolidation requires eliminations.
- **SAMS-3** · 2026-07-30 company event · Q2 management transcript. Management speech: HBM, bit growth, LTAs and foundry capacity. Host displays July 29; company dates results July 30.
- **SAMS-4** · 2026-06-30 share date · Company listing information. Common, preferred and treasury share reconciliation.
- **SAMS-5** · 2026-08-21 · Issuer RNS: share repurchase decision. Employee compensation purpose and planned purchase window; use issuer text, not host AI summary.
- **SAMS-6** · 2026-09-01 · Uploaded model and artifact - User upload: v3_2_master_portfolio(1).xlsx; 005930KS_Samsung_DeepDive.md. Sanctioned 6x multiple and operating vector path.

Original source files are retained in the external version archive.

- **TF-1** - Owner-supplied TrendForce revision, incorporated September 20, 2026: Q3 NAND ASP +18-23% and Q4 +15-20% sequentially. Active selections: +23% and +20%. The exact revised publication/table was not independently retrieved. Source priority is Tier 1 alongside Serenity as instructed by the owner.

- **TF-2** - August 18, 2026, TrendForce NAND supplier research. Samsung NAND revenue estimate $23.06B; SK hynix Group includes Solidigm. The article's aggregate and individual supplier totals do not reconcile; its aggregate and market-share denominator are not used in the model.
- **TF-3** - September 7, 2026, TrendForce DRAM supplier research. Samsung Q2 DRAM revenue estimate $60.98B, including the DRAM product mix. The Samsung allocation uses matched Q2 product estimates in USD, avoiding a new FX conversion.
- **JF-1** - Supplied September 21, 2026: Jeff Pu (@sssjeffpu) and Jukan (@jukan05), "NAND Pricing Breakout Driven by eSSD" summary in the owner's conversation. Quoted Q3 2026 / Q4 2026 / Q1 2027 blended NAND changes: +21% / +17% / +15%, citing TrendForce demand research. Only the Q1 +15% is newly adopted; the owner retains +23% and +20% for 2026. This is an owner-authorized combined case. The original analyst note and exact TrendForce table were not independently retrieved. Enterprise demand commentary is contextual and adds no shipment volume to this update.
- **KB-1** - Owner-supplied image of KB Securities, Jeff Kim / Changmin Lee, dated October 1, 2026: HBM wafer allocation, HBM4 revenue mix, greater-than 100% HBM ASP hypothesis and capital returns. Original image embedded in section 2. Original report URL not supplied; attribution comes from the visible page.
- **TF-HBM** - TrendForce, September 29, 2026: 121% blended HBM ASP forecast,8-Hi/12-Hi tradeoffs and supply constraints. Industry forecast, not Samsung contract guidance.
- **MA-1** - Mirae Asset, May 7, 2026, page 1: 2026 HBM revenue 33.8T forecast. This dated sizing anchor is used only for calibration; its older 2027 price estimate is superseded by MA-2.
- **MA-2** - Mirae Asset, September 7, 2026, pages 1 and 5: 2027 HBM bits+57%, ASP+49%,2026 DRAM USD 246.9B and annual FX 1,443. These reconstruct embedded exposure; the analyst's target and total-company forecast are not imported.
- **SAMS-2E** - Samsung Q2 English earnings deck, July 30, 2026, page 7: Memory revenue 120.8T and HBM/server-demand outlook. Source image embedded in section 2.

Model ownership and remaining gaps: probability 50%, relative cost weights, COGS/opex split, front/packaging cost stresses, yield proxies, price-unit normalization and 2028-2030 persistence are explicitly analytical. No measured Samsung HBM yield or standalone HBM operating margin has been invented. Reconcile new issuer evidence against these controls when available. Source reputation raises the weight of a credible forward signal; it does not turn the future outcome into certainty.

Current model snapshot: Q3 preliminary consolidated earnings anchor updated October 11, 2026 from the October 8 issuer guidance; Q4 2026 onward operating path retained. Architecture evidence remains the October 6 review. Framework 8.0. Analytical Base 83 and Excel Base 83 are unchanged; the old 99 override was removed October 7 and the prior artifact text is now synchronized. Valuation remains 6x forward-four-quarter OP / 6.648649811B economic shares. Eighteen operating quarters and fifteen complete valuation windows remain. Active Q4 2027 ceiling is exactly KRW 641449.0626424863; Q1 2028 is KRW 643911.5502726279. The past Q3 2026 window changes from KRW 474026.2617470095 to 477872.43150801185428835033 59, on F4Q OP 529.53440857130891T. Q3 revenue/OP residuals of 7.413972642427T /4.261972642427T stay unallocated; segment estimates await reconciliation. The 25% persistence diagnostic has 100% overlap adjustment and zero active increment. All price, volume, cost, share and multiple assumptions beyond the Q3 consolidated anchor remain unchanged. Embedded source images are preserved for offline use. Recoverable prior MD/PDF and source hashes are retained in the run archive.